Multivariate Regression model

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2025-09-18

This report interprets the results of two multivariate multiple linear regression (MLR) models fitted to the iris and airquality datasets. The first model predicts Sepal.Length and Petal.Length using Sepal.Width and Petal.Width (iris dataset). The second model predicts Ozone and Solar.R using Wind and Temp (airquality dataset). Each model's results are analyzed in terms of coefficients, statistical significance, goodness-of-fit, and model assumptions.

Model Overview

- **Response Variables**: Sepal.Length (sepal length in cm), Petal.Length (petal length in cm)
- **Predictors**: Sepal.Width (sepal width in cm), Petal.Width (petal width in cm)
- **Sample Size**: 150 observations
- **Model**: Multivariate multiple linear regression (lm(Sepal.Length, Petal.Length ~ Sepal.Width + Petal.Width))

```r
# Load the iris dataset
data(iris)

# Define response variables (Sepal.Length, Petal.Length) and predictors
(Sepal.Width, Petal.Width)
Y <- as.matrix(iris[, c("Sepal.Length", "Petal.Length")])  # Response
variables
X <- iris[, c("Sepal.Width", "Petal.Width")]  # Predictors

# Fit multivariate multiple linear regression model
mlr_model <- lm(Y ~ Sepal.Width + Petal.Width, data = X)

# Summary of the model
summary(mlr_model)
```

```
## Response Sepal.Length :
##
## Call:
## lm(formula = Sepal.Length ~ Sepal.Width + Petal.Width, data = X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -1.2076 -0.2288 -0.0450  0.2266  1.1810
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.45733    0.30919   11.18  < 2e-16 ***
## Sepal.Width  0.39907    0.09111    4.38 2.24e-05 ***
## Petal.Width  0.97213    0.05210   18.66  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4511 on 147 degrees of freedom
## Multiple R-squared:  0.7072, Adjusted R-squared:  0.7033
## F-statistic: 177.6 on 2 and 147 DF,  p-value: < 2.2e-16
##
##
## Response Petal.Length :
##
## Call:
## lm(formula = Petal.Length ~ Sepal.Width + Petal.Width, data = X)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.33753 -0.29251 -0.00989  0.21447  1.24707
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.25816    0.31352   7.203 2.84e-11 ***
## Sepal.Width -0.35503    0.09239  -3.843  0.00018 ***
## Petal.Width  2.15561    0.05283  40.804  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4574 on 147 degrees of freedom
## Multiple R-squared:  0.9338, Adjusted R-squared:  0.9329
## F-statistic:  1036 on 2 and 147 DF,  p-value: < 2.2e-16
```

Summary (Iris)

The model for Sepal.Length explains 70.72% of the variance, with both predictors significant, though Petal.Width has a stronger effect. The model for Petal.Length is exceptionally strong, explaining 93.38% of the variance, with Petal.Width as the dominant predictor and Sepal.Width having a negative effect. The overall model is statistically significant ($p < 0.001$).

Model Overview

- **Response Variables**: Ozone (ozone concentration in ppb), Solar.R (solar radiation in lang)
- **Predictors**: Wind (wind speed in mph), Temp (temperature in °F)
- **Sample Size**: 111 observations (after removing NAs)
- **Model**: Multivariate multiple linear regression (lm(Ozone, Solar.R ~ Wind + Temp))

```r
# Load the airquality dataset
data(airquality)

# Remove rows with missing values to ensure clean data
airquality_clean <- na.omit(airquality)

# Define response variables (Ozone, Solar.R) and predictors (Wind, Temp)
Y <- as.matrix(airquality_clean[, c("Ozone", "Solar.R")])  # Response
variables
X <- airquality_clean[, c("Wind", "Temp")]  # Predictors

# Fit multivariate multiple linear regression model
mlr_model <- lm(Y ~ Wind + Temp, data = X)

# Summary of the model
summary(mlr_model)
```

```
## Response Ozone :
##
## Call:
## lm(formula = Ozone ~ Wind + Temp, data = X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -42.156 -13.216  -3.123  10.598  98.492
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -67.3220    23.6210  -2.850  0.00524 **
## Wind         -3.2948     0.6711  -4.909 3.26e-06 ***
## Temp          1.8276     0.2506   7.294 5.29e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.73 on 108 degrees of freedom
## Multiple R-squared:  0.5814, Adjusted R-squared:  0.5736
## F-statistic: 74.99 on 2 and 108 DF,  p-value: < 2.2e-16
##
##
```

```
## Response Solar.R :
##
## Call:
## lm(formula = Solar.R ~ Wind + Temp, data = X)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -172.71  -54.23    5.31   66.84  188.64
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -49.8135    95.5582  -0.521  0.60323
## Wind          0.6478     2.7151   0.239  0.81187
## Temp          2.9331     1.0136   2.894  0.00461 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 87.9 on 108 degrees of freedom
## Multiple R-squared:  0.08697,    Adjusted R-squared:  0.07006
## F-statistic: 5.144 on 2 and 108 DF,  p-value: 0.007349
```

The model for Ozone explains 58.14% of the variance, with both Wind and Temp as significant predictors, though the fit is moderate. The model for Solar.R is weak, explaining only 8.70% of the variance, with only Temp as a significant predictor. The wide residuals and low R-squared for Solar.R suggest the model is not adequate for this response variable.